Dissertation Weekly Progress Report
Week 2

Date of Last Meeting: 8 October 2026
Date of Next Meeting: 15 October 2026
Current Focus Area: Inflation - Quantum Initial Conditions, Bunch Davies Vacuum

---

As ususal, I work extensively from [1]. Where I have used ideas or results from other sources, these will be directly cited.

1 Isocurvature Constraints

I will first present the most recent results/work on the constraints on isocurvature modes to (as decided in the meeting on 8/10/26) conclude the previous discussion on this topic.

Recent observations have tightly constrained the isocurvature by measuring the CMB. The *Planck* collaboration constrained the contribution to the CMB temperature variance to $1.3\%, 1.7\%,$ and1.7% for CDM, neutrino density, and neutrino velocity isocurvature respectively [2]. Further work has been done to constrain isocurvature in models of new physics that would not fit well to the usual assumptions made in these studies [3].

Since cold dark matter isocurvature (CDI) can source graviational waves (GW) as they enter the horizon [4], improvements in GW detection will allow the use of these to probe and better constrain CDI.

One 'type' of isocurvature that is not as well constrained by the CMB is the *compensated isocurvature mode*. These are opposite-sign perturbations of baryons and dark matter that leave the total matter density unchanged, thus they do not alter gravitational potentials at linear order. They impact the CMB only through a second order effect. Recent work has been done on this to better constrain this form of primordial isocurvature by using BOSS galaxy data [5].

The strong, and still improving, constraints (about zero) on isocurvature in the early universe makes this regime of research possibly too niche for this current work. We will, therefore, leave any further exploration of this subject to further work.

2 Towards the Bunch-Davies Mode Function

2.1 Review of Lagrangian Field Theory

Since our first step to obtainin the Bunch-Davies Mode Function is to arrive at a Klein-Gordon equation for the inflaton [1], it is worth presenting a brief review of Langrangian field theory ([6], Chp. 2), such that we can confidently use these techniques moving forward.

Let $\phi(t, \vec{x})$ be a scalar field. Then we can write a *Lagrangian density* $\mathcal{L}$ as a function of this field and its derivatives $\partial_\mu \phi$. In local field theory, we can express the standard lagrangian L as an integral function of this density:

$$L = \int \mathcal{L}(\phi, \partial_\mu \phi)\, d^3x$$

The action S is defined as the time integral of the Lagrangian, and thus can be written in terms of the Lagrangian density:

$$S = \int L\, dt = \int \mathcal{L}\, d^4x$$

[1] We will encode the time-dependent dynamics during inflation in a scalar field known as the *inflaton* field

By the principle of least action, we want to minimize S to learn how it evolves between fixed configurations within two finite times (t_1, t_2). Thus if S is at an extremum (here, a minimum) we can write $\delta S = 0$. This condition further expands as

$$\begin{aligned} 0 &= \delta S \\ &= \int d^4x \left\{ \frac{\partial \mathcal{L}}{\partial \phi}\delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\delta(\partial_\mu \phi) \right\} \\ &= \int d^4x \left\{ \frac{\partial \mathcal{L}}{\partial \phi}\delta\phi - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) \delta\phi + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\delta\phi \right) \right\} \end{aligned} \tag{2.1}$$

By the Divergence Theorem, we can treat the final of the three terms in 2.1 as a surface integral. Since we already have we assume the initial and final field conditions, we have $\delta\phi(t = t_1, t_2) = 0$. Restricting our analysis to cases where $\delta\phi = 0$ along the spatial boundary, the surface integral then vanishes, thus we are effectively left with:

$$\begin{aligned} &= \int d^4x \left\{ \frac{\partial \mathcal{L}}{\partial \phi}\delta\phi - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) \delta\phi \right\} \\ &= \int d^4x \cdot \delta\phi \left\{ \frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) \right\} \end{aligned}$$

Which clearly gives us the condition:

$$\boxed{\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) - \frac{\partial \mathcal{L}}{\partial \phi} = 0} \tag{2.2}$$

This is the *Euler-Lagrange Equation* for the motion of a scalar field.

The scalar field has kintetic, gradient, and potential energy (density) terms. These can be combined into a Lorentz-invariant "kinetic term"

$$-\frac{1}{2}\eta^{\mu\nu}\partial_\mu\phi\partial_\nu\phi = \frac{1}{2}\dot{\phi}^2 - \frac{1}{2}(\nabla\phi)^2$$

which leads to the Lagrangian:

$$L = \int \mathrm{d}^3x \left[-\frac{1}{2}\eta^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - V(\phi) \right]$$

Applying 2.1 (or by direct varying of the action), we obtain

$$\Box\phi = \eta^{\mu\nu}\partial_\mu\partial_\nu\phi = \frac{\mathrm{d}V}{\mathrm{d}\phi}$$

This is the *Klein-Gordon equation.*

2.2 Equation of Motion for the Inflaton Field

Let us first explicitly define that, from now on, we refer to the scalar field $\phi(t, \vec{x})$ as the *inflaton.*

In Section 2.1, we wrote out the Langrangian for a scalar field in Minkowski space. We now want to generalise this; that is, write this in a way that is applicable to any general curved spacetime. It is obvious that we replace $\eta^{\mu\nu} \to g^{\mu\nu}$, but we also introduce the scaling factor $\sqrt{-g} \coloneqq \sqrt{-\det(g_{\mu\nu})}$ to ensure that the volume element is invariant under a coordinate transformation. This gives us the action for the motion of the inflaton field:

$$S = \int \mathrm{d}^4x\sqrt{-g}\left[-\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - V(\phi)\right] \tag{2.3}$$

Defining $\tilde{\mathcal{L}} \coloneqq \sqrt{-g}\mathcal{L} = \sqrt{-g}\left[-\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - V(\phi)\right]$ we can use 2.1 (with substitution $\mathcal{L} \to \tilde{\mathcal{L}}$) to obtain the Klein-Gordon equation for inflatons:

$$\frac{1}{\sqrt{-g}}\partial_\mu\left(\sqrt{-g}g^{\mu\nu}\partial_\nu\phi\right) = \frac{\mathrm{d}V}{\mathrm{d}\phi} \tag{2.4}$$

I have included the derviation of this in A.

2.3 Mukhanov-Sasaki Equation

We will for now focus only on scalar pertubations and thus take the metric of the form:

$$g_{\mu\nu} = a^2(\eta)\begin{pmatrix} -(1+2A) & \partial_j B \\ \partial_i B & \delta_{ij} \end{pmatrix} \quad \text{and} \quad g^{\mu\nu} = a^{-2}(\eta)\begin{pmatrix} -(1-2A) & \partial^j B \\ \partial^i B & \delta^{ij} \end{pmatrix}$$

Then $\sqrt{-g} = a^4(1+A)$.
In this metric, 2.4 gives

$$\delta\phi'' + 2\mathcal{H}\delta\phi' - \nabla^2\delta\phi = (A' + \nabla^2 B)\bar{\phi}' - 2a^2V_{,\phi}A - a^2V_{,\phi\phi}\delta\phi$$

where we introduce the notation $V_{,\phi} = \mathrm{d}V/\mathrm{d}\phi$, $V_{,\phi\phi} = \mathrm{d}^2V/\mathrm{d}\phi^2$ and perturb the field in the usual way $\phi = \bar{\phi} + \delta\phi$.

We now are motivated to write this equation in a way that is solely in terms of the inflaton field fluctuation $\delta\phi$. This can be done using the Einstein equations. It is, however, a notably tedious procedure and I will not include the derivation here.

First we will introduce some new terminology,

Definition 2.1 (*Slow-roll Parameters*).

$$\varepsilon \coloneqq -\frac{\dot{H}}{H^2} = 1 - \frac{\mathcal{H}'}{\mathcal{H}^2} = 4\pi G\frac{(\bar{\phi}')^2}{\mathcal{H}^2} \tag{2.5}$$

$$\delta \coloneqq -\frac{\ddot{\bar{\phi}}}{H\dot{\bar{\phi}}} = 1 - \frac{\bar{\phi}''}{\mathcal{H}\bar{\phi}'} \tag{2.6}$$

Then using this notation, and the constraint eqautions that do not involve time derivatives of the metric perturbations, we arrive at the relations

$$A = \varepsilon\frac{\mathcal{H}}{\bar{\phi}'}\delta\phi \tag{2.7}$$

$$\nabla^2 B = -\varepsilon\frac{\mathcal{H}}{\bar{\phi}'}\left(\delta\phi' + (\delta - \varepsilon)\mathcal{H}\delta\phi\right) \tag{2.8}$$

This allows us to obtain a closed evolution equation for the inflaton fluctuations:

$$\delta\phi'' + 2\mathcal{H}\delta\phi' - \nabla^2\delta\phi = \left[2\varepsilon(3 + \varepsilon - 2\delta) - \frac{a^2V_{,\phi\phi}}{\mathcal{H}^2}\right]\mathcal{H}^2\delta\phi$$

We note that the mixing with the metric fluctuation only contributes to the *effective mass* of the inflaton fluctuations.

If we then use 2.4 to write $V_{,\phi\phi}$ in terms of ϵ and δ, and make the substitution $f \coloneqq a\delta\phi$, we arrive at

$$\delta\phi'' + 2\mathcal{H}\delta\phi' = \frac{1}{a}\left(f'' - (2-\varepsilon)\mathcal{H}^2 f\right)$$

.

We can collect terms, and introducing the notation $z = a\bar{\phi}'/\mathcal{H}$ we can write this equation it its most famous form; the *Mukhanov-Sasaki Equation*...

$$\boxed{f'' + \left(k^2 - \frac{z''}{z}\right) f = 0} \tag{2.9}$$

There are a few points worth making about this equation:

- it makes no slow-roll approximation
- the coupling of the metric and matter fluctuations are contained within it
- it is valid on all scales

All of this makes it a rather incredible equation, and it shall be the cornerstone of our work going forward.

2.4 Superhorizon Freezout

Recall the equation for a simple harmonic oscillator (SHO): $\ddot{x} + \omega^2 \dot{x} = 0$. Thus we notice that 2.9 is exactly a SHO with a (conformal) time-dependent frequency $\omega_f^2(\eta, k) = k^2 - \frac{z''}{z}$.

During slow-roll inflation [2], we have that H[3] and $\frac{d\bar{\phi}}{dt}$ are approximately constant. Recalling the relationships $\mathcal{H} = aH$ and $\bar{\phi}' = \mathrm{d}\bar{\phi}/\mathrm{d}\eta = a(t)\mathrm{d}\bar{\phi}/\mathrm{d}t$ we also ahve $\bar{\phi}'/\mathcal{H}$ is approximately constant. Furthermore

$$\frac{z''}{z} \approx \frac{a''}{a} \approx 2\mathcal{H}^2$$

where we use the fact $a' = a^2 H$. [4]

Thus the inverse of z''/z is a measure of the comoving hubble radius, or *horizon*, $\mathcal{H}^{-1}$.

We then inspect the *subhorizon limit*, that is $k^2 \gg |z''/z|$: Then 2.9 reduces to

$$f'' + k^2 f = 0$$

. Such a limit applies at early times when $\mathcal{H}^{-1}$ is large and all modes are inside the horizon. This is now a SHO with a fixed frequency $\omega = k$. As usual, this has solutions $f \propto e^{\pm ik\eta}$.

Secondly, we look at the *superhorizon limit*, that is $k^2 \ll |z''/z|$. In this limit,

$$f'' - \frac{z''}{z} f = 0$$

. As the comoving horizon shrinks, microscopic scales are stretched outside of the constant Hubble radius H^-1, becoming macroscopic cosmological fluctuations. The equation in this limit has a growing mode solution and a decaying mode solution, $f \propto z$ and $f \propto z^{-2}$ respectively.

[2] I will include a deeper introduction to inflation in a future report, for now I want to focus on the ideas presented here.

[3] In this font, I take it to refer to the Hubble parameter in cosmic time t.

[4] If it is not clear, we refer to derivatives w.r.t. conformal time with a prime $'$ and those w.r.t. cosmic time with a dot $\dot{}$.

The growing mode corresponds to a frozen curvature perturbation, as will be shown. This is called the *freezeout*.

Recall the curvature perturbation in the comoving gauge, from a previous discussion of perutbation theory [5]:

$$\mathcal{R} = -C + \frac{1}{3}\nabla^2 E - \mathcal{H}(v + B).$$

In the spatially flat gauge this becomes

$$\mathcal{R} = -\mathcal{H}(v + B)$$

where we defined v by $\delta T^i_0 = -(\bar{\rho} + \bar{P})\partial^i v$.

The energy-momentum tensor for a scalar field is given by $T_{\mu\nu} = \partial_m u \partial_n u \phi - g_{\mu\nu}(\frac{1}{2} g^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi + V(\phi))$ and thus

$$\begin{aligned}\delta T^i_0 = g^{i\mu}\partial_\mu\phi\partial_0\phi &= g^{i0}(\bar{\phi}')^2 + g^{ij}\bar{\phi}'\partial_j\delta\phi \\ &= a^{-2}(\bar{\phi}')^2\partial^i(B + \delta\phi/\bar{\phi}').\end{aligned}$$

From this we read off that $\bar{\rho} + \bar{P} = a^{-2}(\bar{\phi}')^2$ and thus $v = -B - \delta\phi/\bar{\phi}' \implies v + B = -\delta\phi/\bar{\phi}'$. Combining these results gives us the form of $\mathcal{R}$ as follows,

$$\mathcal{R} = \frac{\mathcal{H}}{\bar{\phi}'}\delta\phi = \frac{f}{z}$$

Which tends to a constant in the limit $k \ll \mathcal{H}$ since we have $f \propto z$.

We will want to plot this, but solving 2.9 requires picking initial conditions. The next section will motivate our eventual choice, and as such we delay the plotting until then. However, I will provide a figure to show what we expect to see:

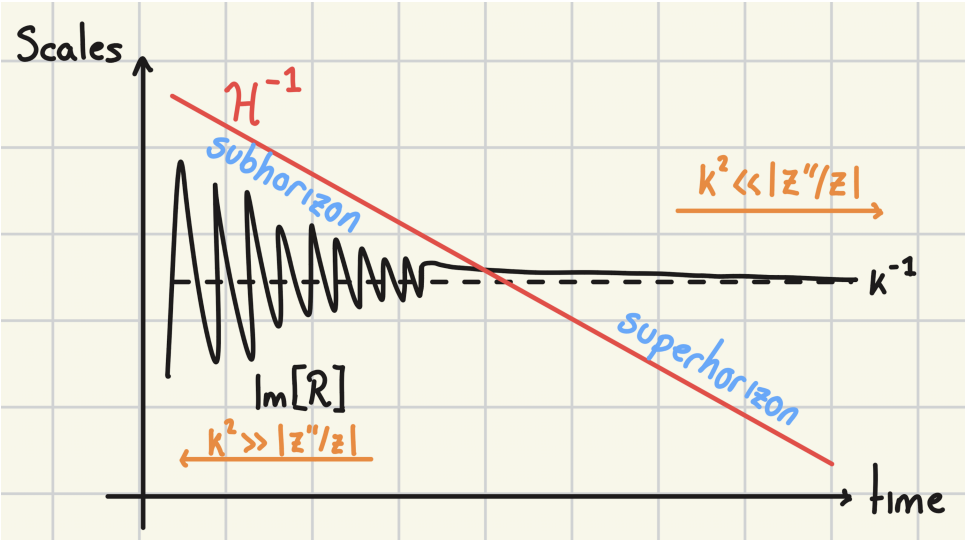


Figure 1: Qualitative illustration of how we expect the curvature fluctuation to decay from an oscilatting mode to a constant mode, the process of *freezeout*.

2.5 Inflationary Vacuum Fluctuations

We will continue by working in *de Sitter spacetime*. That is (for our current purposes [6]) the spacetime given by the metric

$$ds^2 = -dt^2 + e^{2Ht}dx^2.$$

[5] For those following on my website, this was defined in a rough notes section prior to my starting of these reports.

[6] I would like to include a more formal mathematical definition at some later stage, if it feels relevant

This is a good approximation if the inflationary background when $\epsilon \neq 0$ is small. Thus pairing this with the slow-roll approximation, we have $H =$ const and $a = -(H\eta)^{-1}$, and 2.9 becomes

$$f_{\mathbf{k}}'' + \left(k^2 - \frac{2}{\eta^2}\right) f_{\mathbf{k}} = 0 \tag{2.10}$$

To be continued....

2.6 Questions for this weeks meeting

As usual, I will write down some questions that I hope will be answered in the next meeting:

- How do we proceed make sure my dissertation doesn't just become a review article of inflation?
  - I am eager to try to get some novel result in my dissertation if possible, how should I search this gap in the literature out?
  - How do I depart from the textbook and to research-level, without going too niche?
- Is this roughly the right amount to be doing a week? Should I be doing more?

A Derivation of the Inflaton Klein-Gordon Equation

Although one can vary the inflaton field action directly, we present the generalisation of the classical Euler-Lagrange equation to a general spacetime quantified by some metric $g^{\mu\nu}$.

We can clearly make the replacement $\tilde{\mathcal{L}} \to \tilde{\mathcal{L}} = \sqrt{-g}\mathcal{L}$ which gives us (from 2.1) the form

$$\partial_\mu \left(\frac{\partial \tilde{\mathcal{L}}}{\partial(\partial_\mu \phi)}\right) - \frac{\partial \tilde{\mathcal{L}}}{\partial \phi} = 0$$

.

Using the product rule we get the final form of

$$\partial_\mu \left(\sqrt{-g}\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\right) - \sqrt{-g}\frac{\partial \mathcal{L}}{\partial \phi} = 0$$

Taking $\mathcal{L} = -\frac{1}{2}g^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi - V(\phi)$ we calculate

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = -\frac{1}{2}g^{\alpha\beta}\left(\delta^\mu_\alpha \partial_\beta\phi + \partial_\alpha\phi\,\delta^\mu_\beta\right) = -\frac{1}{2}g^{\mu\beta}\partial_\beta\phi - \frac{1}{2}g^{\alpha\mu}\partial_\alpha\phi = -g^{\mu\nu}\partial_\nu\phi$$

$$\frac{\partial \mathcal{L}}{\partial \phi} = -\frac{dV}{d\phi}$$

Then by substitution into our modified Euler-Lagrange equation we get

$$\boxed{\frac{1}{\sqrt{-g}}\partial_\mu\left(\sqrt{-g}\,g^{\mu\nu}\partial_\nu\phi\right) = \frac{dV}{d\phi}}$$

References

[1] Daniel Baumann. *Cosmology*. Cambridge Aspire website. ISBN: 9781108937092. June 30, 2022. DOI: 10.1017/9781108937092. URL: https://www.cambridge.org/highereducation/books/cosmology/53783DD7B3CB15E2E37ADFBC0C1B930F (visited on 10/11/2026).

[2] Planck Collaboration et al. "Planck 2018 results. X. Constraints on inflation". In: *Astronomy & Astrophysics* 641 (Sept. 2020), A10. ISSN: 0004-6361, 1432-0746. DOI: 10.1051/0004-6361/201833887. arXiv: 1807.06211[astro-ph.CO]. URL: http://arxiv.org/abs/1807.06211 (visited on 10/08/2026).

[3] Matthew R. Buckley et al. *General Constraints on Isocurvature from the CMB and Ly-α Forest*. version: 2. Apr. 1, 2025. DOI: 10.48550/arXiv.2502.20434. arXiv: 2502.20434[astro-ph.CO]. URL: http://arxiv.org/abs/2502.20434 (visited on 10/08/2026).

[4] Guillem Domènech, Samuel Passaglia, and Sébastien Renaux-Petel. "Gravitational waves from dark matter isocurvature". In: *Journal of Cosmology and Astroparticle Physics* 2022.3 (Mar. 1, 2022), p. 023. ISSN: 1475-7516. DOI: 10.1088/1475-7516/2022/03/023. arXiv: 2112.10163[astro-ph.CO]. URL: http://arxiv.org/abs/2112.10163 (visited on 10/06/2026).

[5] Alexandre Barreira. *Constraints on compensated isocurvature perturbations from BOSS DR12 galaxy data*. Feb. 3, 2023. DOI: 10.48550/arXiv.2302.01927. arXiv: 2302.01927[astro-ph.CO]. URL: http://arxiv.org/abs/2302.01927 (visited on 10/15/2026).

[6] Michael E. Peskin. *An Introduction To Quantum Field Theory*. Boca Raton: CRC Press, May 4, 2018. 866 pp. ISBN: 978-0-429-50355-9. DOI: 10.1201/9780429503559.